

CSE340: Computer Architecture

Question 1, 2, 3, 4, 7, 8, 9, 10 are theory questions check handout/slide for solutions.

Question - 5:

$$\text{Cost per die} = \frac{\text{Cost per wafer}}{\text{Dies per wafer} \times \text{yield}}$$

Observe this formula, why do we multiply Dies per wafer with yield ?

Solution: We must calculate the cost per die based on the number of

Working(functional) dies produced as we can not sell the defective ones.

Dies per wafer is the total number of die produced and yield is the % of working dies. Therefore, multiply both we get the number of working dies produced.

Question - 6:

A processor executes a program, which has a total of 38×10^9 instructions, categorized as follows:

Instruction Type	Instruction Count	Cycle per instruction
R-type	35% of total instructions	2
I-type	25% of total instructions	3
SB-type	30% of total instructions	4
U-type	10% of total instructions	1

- Find the average CPI of the program.
- If the CPU clock frequency is 2.5 GHz, calculate the execution time of the program.

Solution:

Instruction Type	Instruction Count	Cycle per instruction	Instruction Count
R-type	35% of total instructions	2	$38 \times 10^9 \times 0.35 = 133 \times 10^8$
I-type	25% of total instructions	3	$38 \times 10^9 \times 0.25 = 95 \times 10^8$
SB-type	30% of total instructions	4	$38 \times 10^9 \times 0.30 = 114 \times 10^8$
U-type	10% of total instructions	1	$38 \times 10^9 \times 0.10 = 38 \times 10^8$

a) Avg cpi = $[(133 \times 10^8 \times 2) + (95 \times 10^8 \times 3) + (114 \times 10^8 \times 4) + (38 \times 10^8 \times 1)] / 38 \times 10^9 = \mathbf{2.75}$

b) Execution time = $(IC \times CPI) / F = (38 \times 10^9 \times 2.75) / 2.5 \times 10^9 = \mathbf{41.8s}$

Question - 8:

Can you relate/connect Amdahl's law with any of the design principles, mentioned below?

Solution: Make the common case faster.

Question - 11:

Consider a computer running a program that requires 200s, with 70s spent executing R-type instructions, 85s executed I/S type instructions, and 45s spent executing branch instructions.

Can the total time be reduced by 20% by reducing only the time for branch instructions? If yes, **Determine** the improvement factor.

Solution:

Total Time = 200s

New Total Time, $T_{imp} = 200 \times 0.8 = 160s$ [time be reduced by 20% meaning 80% remains]

$T_{aff} = 45s$

$T_{unaff} = 200 - 45 = 155$

$$T_{imp} = T_{aff}/n + T_{unaff}$$

$$160 = 45/n + 155$$

$$45/n = 5$$

$$n = 9$$

Yes, the total time be reduced by 20% by reducing only the time for branch instructions. Improvement Factor = **9**

Question - 12:

In the context of evaluating CPU performance, three key factors are considered:

1. Instruction Count
2. CPI (Cycles Per Instruction)
3. Clock Rate

Which of these factors are affected by the choice of programming language, and in what way?

Solution:

Instruction Count and **CPI**.

Question - 13:

Consider the following table which includes information regarding three different processors A, B and C. Now, find the Execution Time in seconds for different programs executed in Processor A, B and C based on the given information.

Processor	MIPS	Instruction Count
A	500	2×10^{10}
B	200	3×10^9
C	1000	1.5×10^{10}

Solution:

$$\text{MIPS}_A = \text{IC}_A / \text{ET}_A \times 10^6$$

$$500 = 2 \times 10^{10} / \text{ET}_A \times 10^6$$

$$\text{ET}_A = 2 \times 10^4 / 500$$

$$= 40\text{s}$$

$$\text{MIPS}_B = \text{IC}_B / \text{ET}_B \times 10^6$$

$$500 = 3 \times 10^9 / \text{ET}_B \times 10^6$$

$$\text{ET}_B = 3 \times 10^3 / 200$$

$$= 15\text{s}$$

$$\text{MIPS}_C = \text{IC}_C / \text{ET}_C \times 10^6$$

$$500 = 1.5 \times 10^{10} / \text{ET}_C \times 10^6$$

$$\text{ET}_C = 1.5 \times 10^4 / 1000$$

$$= 15\text{s}$$